

MATH 345 Skeleton Notes

Unit 2: Anatomy of a linear map

Section 2.6: Coding recap

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Linked notebook

Run [Lab U2: Understanding a Linear Map](#). The main sequence practices projection, reachable outputs, row reduction, plane intersections, difference matrices, rank, null vectors, redundant features, determinants, and inverse checks.

For a quick reference on arrays, matrix products, numerical rank, determinants, solves, row reduction, and null spaces, see the programming appendix sections NumPy Arrays, Vectors, Matrices, and Shapes, Elementwise Arithmetic Versus Linear Algebra, NumPy Linear Algebra Commands, and SymPy Symbolic Computation.

Review activities

Activity: A sensor that drops height (U2-LO3, U2-LO6)

A sensor records only horizontal position. It keeps the first two coordinates and drops height.

```
import numpy as np

P = np.array([
    [1., 0., 0.],
    [0., 1., 0.],
])

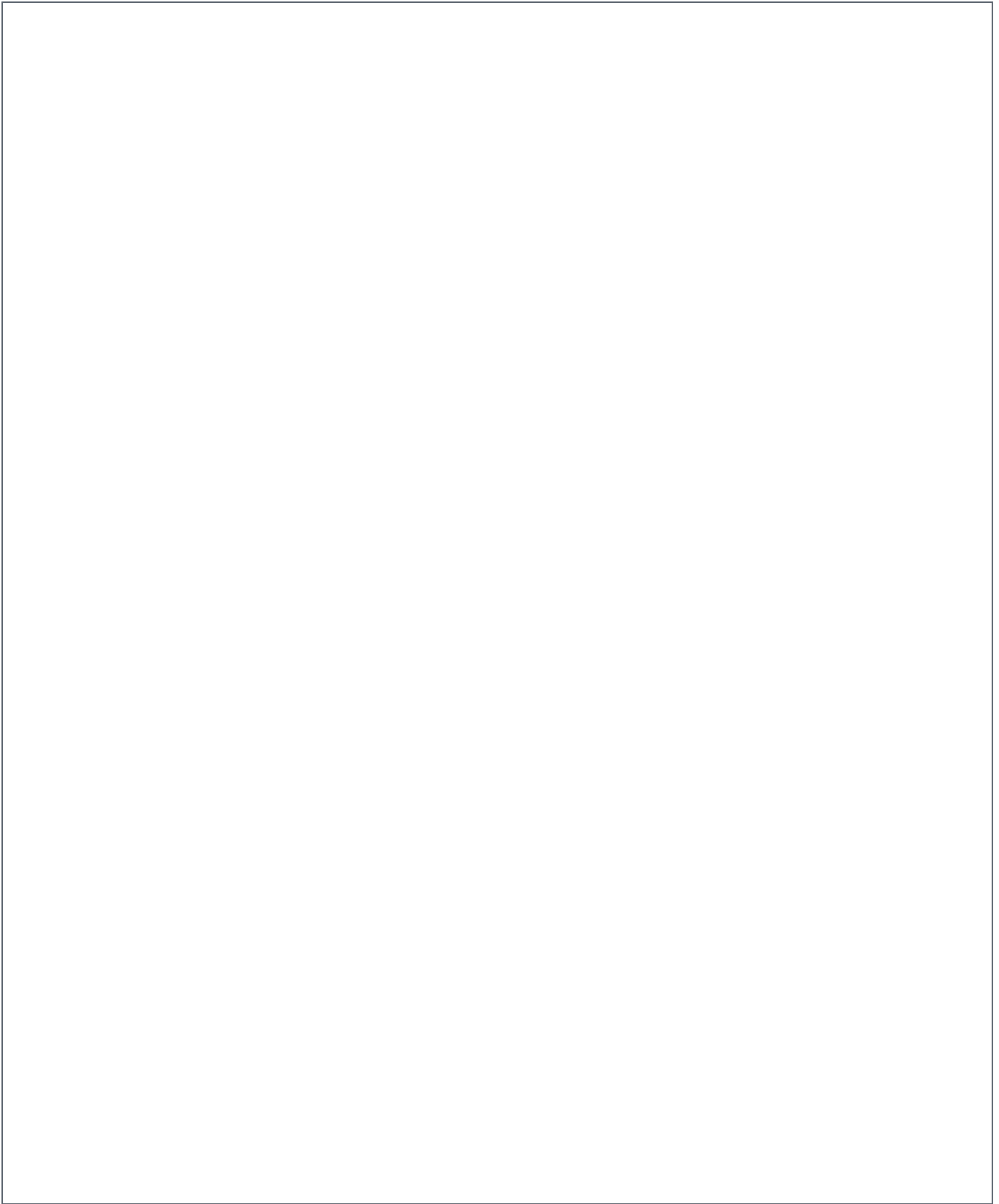
x = np.array([2., 1., 5.])
z = np.array([0., 0., 1.])

P @ x, P @ (x + 6*z), P @ z
```

Output:

```
(array([2., 1.]), array([2., 1.]), array([0., 0.]))
```

1. What does P do to an input vector?
2. Why do Px and $P(x + 6z)$ agree?
3. What does $Pz = 0$ say about the direction z ?
4. Why can the original input not be recovered uniquely from the output?



Activity: Reading a null-space basis in code (U2-LO3, U2-LO6)

The following code asks SymPy for a basis of a null space. Recall from Using a SymPy null-space vector (U2-LO3, U2-LO4, U2-LO6) how to select a vector from the returned list and multiply it by the matrix.

```
import sympy as sp

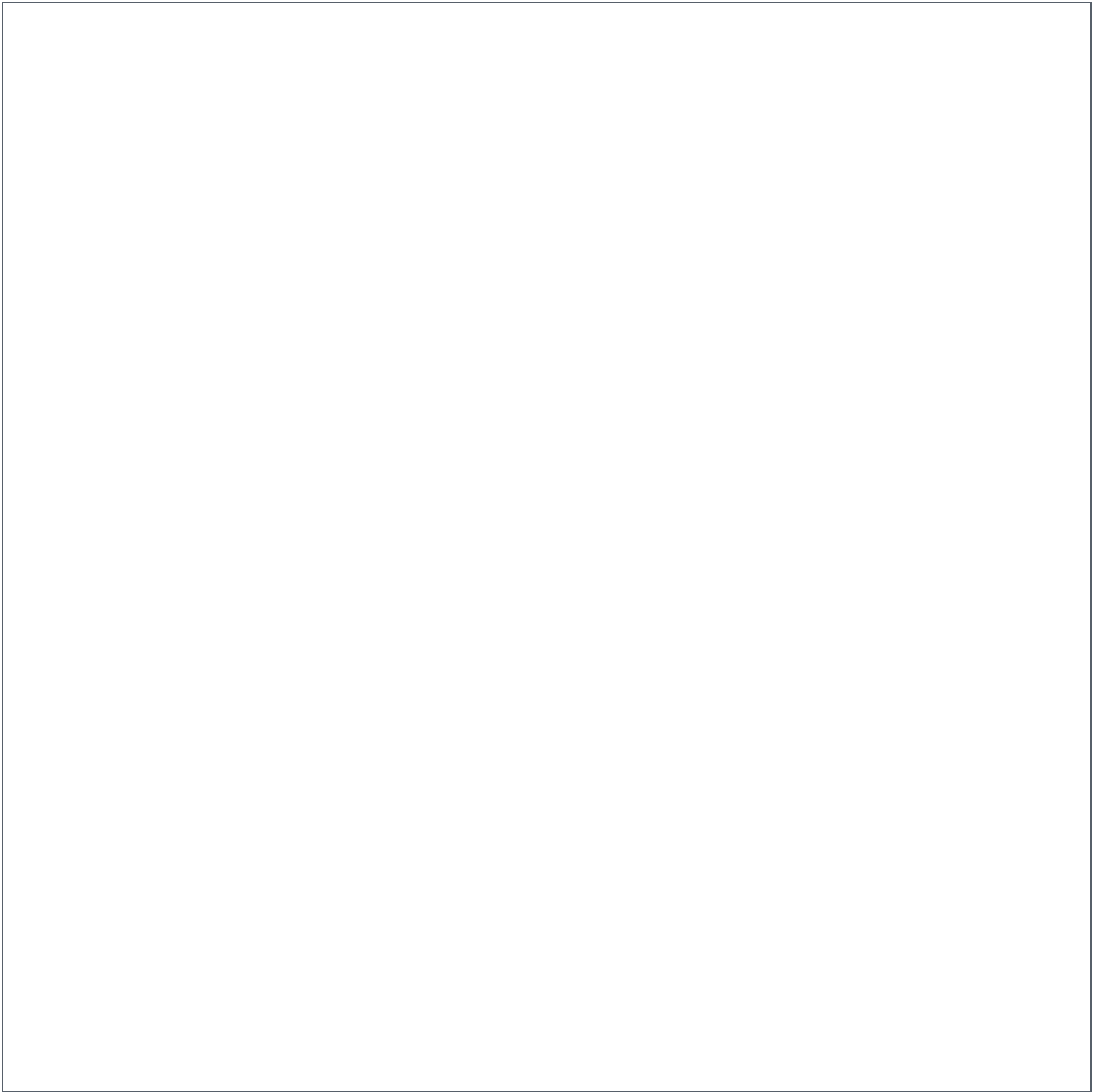
A = sp.Matrix([
    [1, 2, 3],
    [2, 4, 6],
    [0, 1, 1],
])

A.nullspace()
```

Output:

```
[Matrix([
[-1],
[-1],
[ 1]])]
```

1. What kind of object does `nullspace()` return?
2. Is the displayed vector an input direction or an output direction?
3. Verify from the code output that $\mathbf{z} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$ is in $\text{null}(A)$.
4. If $A\mathbf{x} = \mathbf{y}$, what is $A(\mathbf{x} + t\mathbf{z})$?



Activity: Reading an augmented rref output (U2-LO1, U2-LO2, U2-LO3)

Recall from Reading row reduction in code (U2-LO1, U2-LO2) how to read row-reduction output. For unpacking the result and selecting entries using zero-based indices, see Selecting original pivot columns in SymPy (U2-LO3, U2-LO4).

The code row-reduces the augmented matrix for trying to solve

$$Ax = b, \quad A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

```
import sympy as sp

# Augmented matrix [A | b]
M = sp.Matrix([
    [1, 0, 1],
    [0, 1, 2],
    [1, 1, 4],
])

M.rref()
```

Output:

```
(Matrix([
[1, 0, 0],
[0, 1, 0],
[0, 0, 1]]), (0, 1, 2))
```

Read the displayed matrix as

$$\left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right].$$

Activity: Reading an augmented rref output (U2-LO1, U2-LO2, U2-LO3)
(continued)

1. What does the last row represent?
2. Is $\mathbf{b}$ reachable as an output of A ?
3. What does the pivot to the right of the vertical line mean?

Activity: Redundant exam-score features in code (U2-LO3, U2-LO6)

Use the exam-parts-and-total example from Independence, bases, and dimension. Rows represent students. The four columns record scores on part 1, part 2, part 3, and the exam total, all measured in points.

```
import numpy as np

X = np.array([
    [8., 6., 4., 18.],
    [5., 9., 7., 21.],
    [7., 4., 9., 20.],
    [6., 8., 5., 19.],
])

z = np.array([1., 1., 1., -1.])
c = np.array([2., 3., 1., 0.])
c_alt = np.array([1., 2., 0., 1.])

rank = np.linalg.matrix_rank(X)

X @ z, rank, X @ c, X @ c_alt
```

Output:

```
(array([0., 0., 0., 0.]),
 3,
 array([38., 44., 35., 41.]),
 array([38., 44., 35., 41.]))
```

1. What does $Xz = 0$ say about the four feature columns?
2. What does $\text{rank}(X) = 3$ say about the four features?
3. Why do Xc and Xc_{alt} agree?
4. What warning does this give about interpreting individual coefficients?



Activity: Debugging a column-space basis from code (U2-LO3, U2-LO4)

A student computes an rref and pivot columns. Recall the column slicing and list comprehension from Selecting original pivot columns in SymPy (U2-LO3, U2-LO4).

```
import sympy as sp

A = sp.Matrix([
    [1, 2, 3],
    [2, 4, 6],
    [0, 1, 1],
])

R, pivots = A.rref()
R, pivots
```

Output:

```
(Matrix([
[1, 0, 1],
[0, 1, 1],
[0, 0, 0]]), (0, 1))
```

The student says: “Columns 1 and 2 of R are a basis for $\text{col}(A)$.”

1. What do the pivot indices $(0, 1)$ mean in mathematical column numbering?
2. What is wrong with using columns of R as a basis for $\text{col}(A)$?
3. Which columns should be used instead?
4. Write a basis for $\text{col}(A)$.



Activity: Determinant and inverse diagnostics in code (U2-LO3, U2-LO5, U2-LO7)

The following code checks determinant and rank for two square matrices.

```
import numpy as np

A = np.array([
    [1., 1.],
    [0., 1.],
])

B = np.array([
    [1., 0.],
    [0., 0.],
])

detA = float(np.linalg.det(A))
detB = float(np.linalg.det(B))

rankA = np.linalg.matrix_rank(A)
rankB = np.linalg.matrix_rank(B)

detA, rankA, detB, rankB
```

Output:

```
(1.0, 2, 0.0, 1)
```

1. Which matrix is invertible?
2. Which matrix forgets a nonzero input direction?
3. What do determinant and rank each say?
4. Should $Bx = \mathbf{b}$ have a unique solution for every $\mathbf{b} \in \mathbb{R}^2$?

